

AI4Climate: Collection of Machine Learning Tasks and Datasets for Tackling Climate Change

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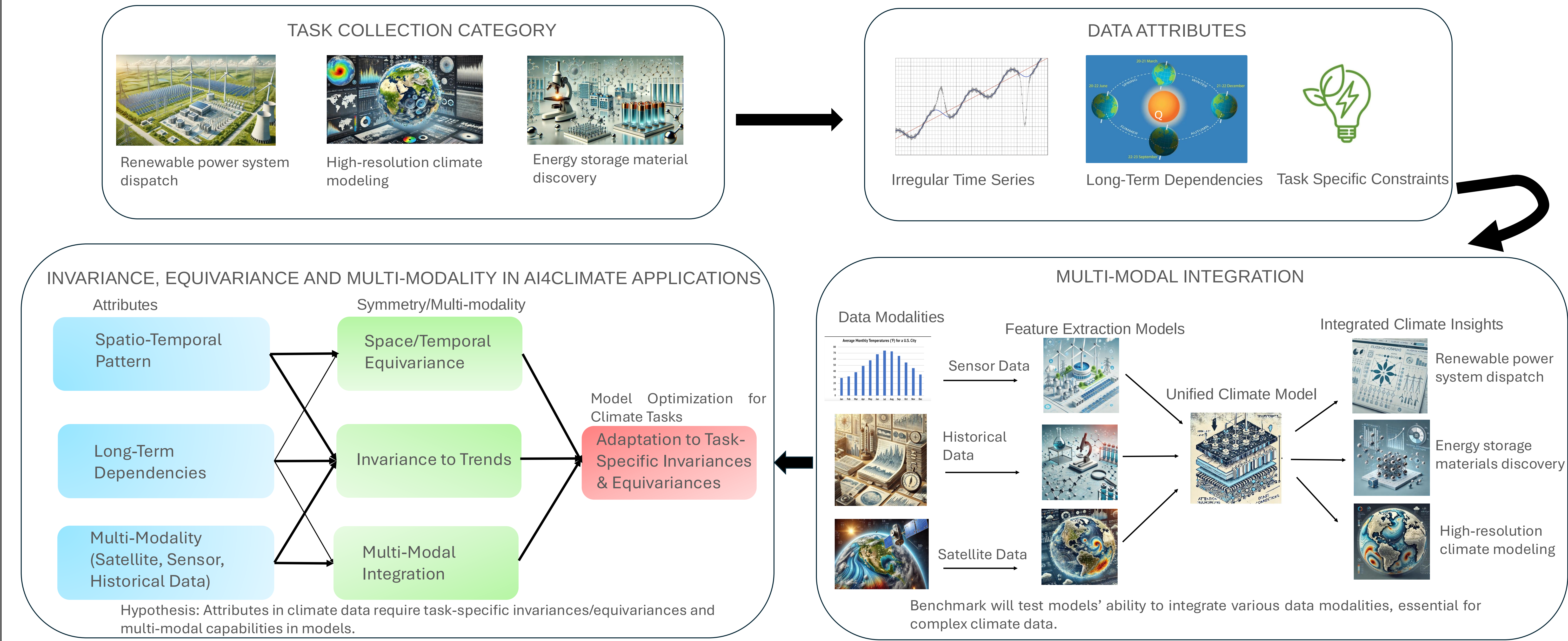
INTRODUCTION

In recent years, foundation models have transformed the field of machine learning, enabling multi-task approaches that perform well across a variety of applications with minimal task-specific data (zero-shot and few-shot learning). However, these models have not achieved similar success in climate and energy domains, where the tasks present unique challenges such as long-term dependencies, spatio-temporal dynamics, and adherence to physical constraints. Mainstream ML applications like computer vision and natural language processing rarely require models to account for these complexities, resulting in poor performance of current foundation models when applied to climate-related tasks. To address this gap, we propose a standardized AI-for-Climate benchmark designed specifically for the challenges of climate science and renewable energy, providing ML practitioners with a comprehensive tool to develop and evaluate models capable of addressing the specific demands of these critical applications.

OBJECTIVES

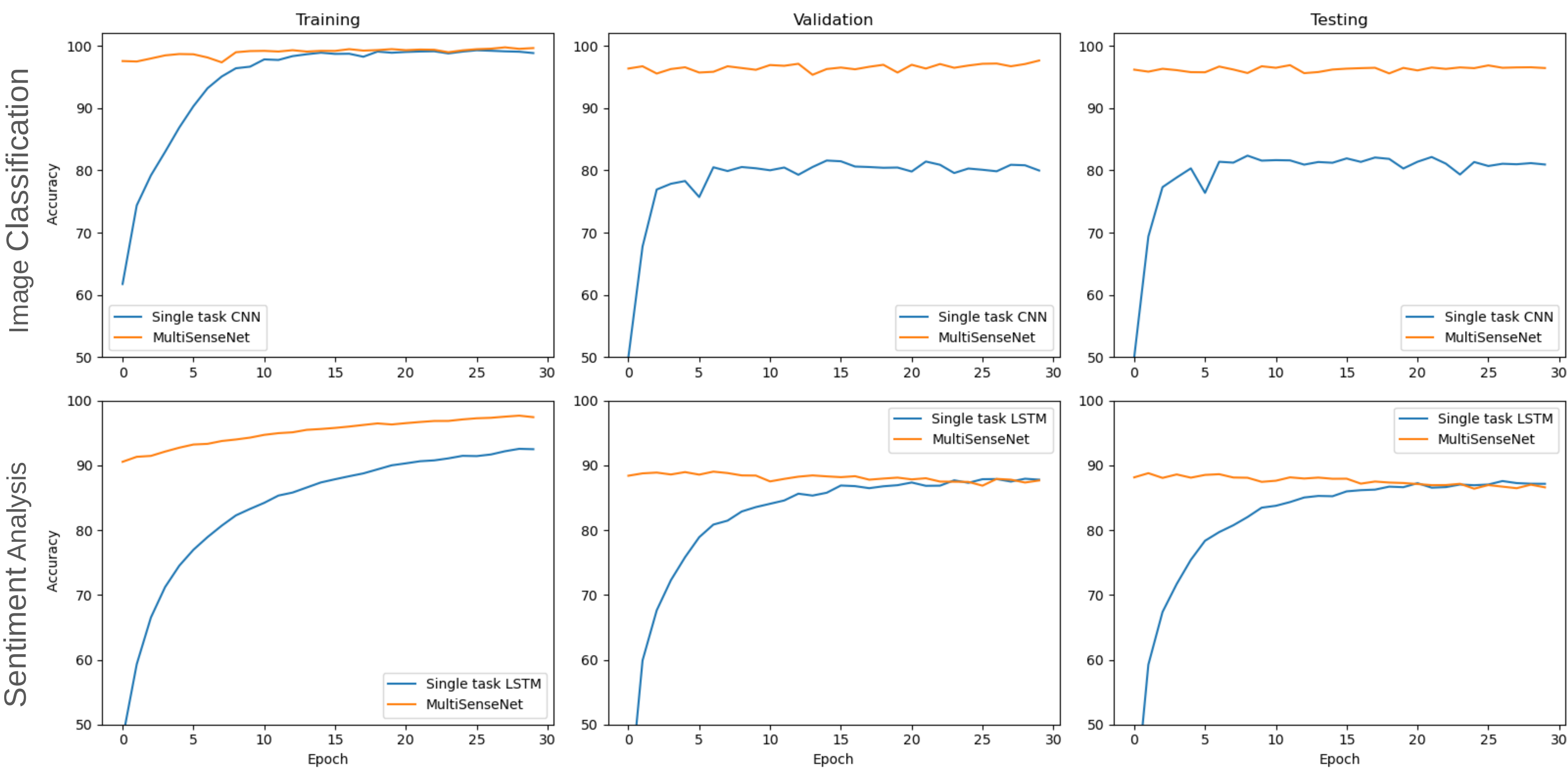
- ❖ Develop a standardized collection of AI-for-climate applications with specific tasks relevant to climate science.
- ❖ Address task-specific challenges like spatio-temporal dependencies, physical constraints, non-stationary data, and multi-modality.
- ❖ Provide ML practitioners with tools to design models that perform well on climate tasks.

METHODOLOGY AND BENCHMARK DESIGN



PREVIOUS WORK (MULTI-SENSE-NET)

Performance of single task models vs MultiSenseNet



POTENTIAL IMPACT

- ❖ Improved model performance on climate tasks
- ❖ Accelerated progress in renewable energy integration and climate science
- ❖ Foundation for future ML research in complex domains



REFERENCES

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2. Higgins, I., Racanière, S., & Rezende, D. (2022). Symmetry-Based Representations for Artificial and Biological General Intelligence. Frontiers in Computational Neuroscience, 16, Article 836498.

EVALUATION METRICS

- ❖ Spatio-Temporal Accuracy
- ❖ Adherence to Physical Constraints
- ❖ Multi-Modal Data Handling

FUTURE WORK

- ❖ Expanding Task Diversity
- ❖ Incorporating Real-Time Adaptation
- ❖ Cross-Domain Model Transferability

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